Uniwersytet SWPS

COURSE SYLLABUS

Developer Tools

ECTS: **3**	Number of Contact Hours: **24**	Student Workload: **75 hours**

FACULTY
Wydział Projektowania w Warszawie

FIELD OF STUDY
Computer Science

STUDY MODE	STUDY PROFILE	LANGUAGE
Full Time	**Practical**	**English**
YEAR OF STUDY	ACADEMIC YEAR	SEMESTER
1	**2026/2027**	**Winter**

Course components

Developer Tools	Laboratory	24

Curriculum Content

The course aims to introduce students to essential development and working tools. During the classes, students will become familiar with the Linux operating system and basic Bash shell commands, enabling them to navigate such systems freely, install software packages, as well as read and edit files. In the latter part of the course, students will learn the fundamentals of version control systems using Git as a practical example.

Course Contents

- **Introduction to Linux**
- **Basic command-line tools**
- **Reading and editing files**
- **Text editors**
- **Bash shell scripting**
- **Integrated Development Environments (IDEs)**
- **Basic Git commands**
- **Introduction to containerization**

Faculty members

Primary faculty
dr Maksymilian Bujok
mbujok@swps.edu.pl

Laboratory
mgr inż. Krzysztof Ropiak kropiak@swps.edu.pl

Learning Outcomes

Knowledge
Has basic knowledge of the programmer's work environment, in particular the Linux operating system. Understands the concepts and benefits associated with application versioning and virtualization using the example of containerization. (INF1_W01)

Skills
Is able to use basic shell commands, write and run a simple Bash script. Knows how to create and manage a project repository. Knows how to create a configuration file to create and run a container (of low complexity).)INF1_U05, INF1_U15) Knows how to create a configuration file to create and run a container (of low complexity). (INF1_U14)

Social Competences
The acquired knowledge allows for searching for sources for further development. (INF1_K01) The acquired knowledge and skills allow for a critical approach to received content, as well as source verification. (INF1_K02)

Student workload distribution

Independent student work	Independent acquisition and processing of knowledge	26
Independent student work	Preparation for assessments and examinations	25
Number of contact hours	Laboratory	24
	Total student workload	**75**

Learning Outcome Assessment Methods

Achieving the minimum score specified in the table is a requirement for passing each task. To pass the course, it is necessary to complete all tasks outlined in the syllabus.

	Minimum number of points required to pass the task	Maximum number of points available for the task

Mixed-Format Test
Laboratory Class Regulations
- **Format and Frequency:** Laboratory sessions take place during every class.
- **Assessment Method:** Students receive graded tasks to complete during class time.
- **Scope of Material:** Students are required to cover all material from all class sessions.
- **Scoring System:**
  - **Maximum score:** 100 points
  - **Minimum score (passing grade):** 51 points
	51	100

Grading Criteria

Very Good	5,0	from 90,50% to 100%
Good Plus	4,5	from 80,50% to 90,49%
Good	4,0	from 70,50% to 80,49%
Satisfactory Plus	3,5	from 60,50% to 70,49%
Satisfactory	3,0	from 50,50% to 60,49%

Fail	2,0	below 50,50%

The final grade is calculated based on the ratio of the sum of points obtained to the sum of points possible, expressed as a percentage.

Attendance requirements

§ 22 Regulaminu Studiów, ust.4 Zakres dopuszczalnych nieobecności określa sylabus przedmiotu. W przypadku nieokreślenia w sylabusie przedmiotu zakresu dopuszczalnych nieobecności, przyjmuje się, że zakres dopuszczalnej nieobecności wynosi nie więcej niż 20%.

Bibliography

Required	Classic Shell Scripting. Hidden Commands that Unlock the Power of Unix S. Chacon, Pro Git – professional version control